

Why Do Black Holes Exist?

A Multi-Messenger Research Plan to Test Penrose's Prediction of Gravitational Collapse

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Abstract—Black holes exist because sufficiently massive stars, once nuclear burning can no longer support them, undergo gravitational collapse, and because—as Sir Roger Penrose proved in 1965 and for which he shared the 2020 Nobel Prize in Physics—such collapse generically produces spacetime singularities hidden within horizons. Yet the empirical status of the three pillars underlying this statement remains only partially quantitative. This research plan formulates three falsifiable hypotheses: (H1) that collapse remnants are Kerr black holes satisfying the no-hair property; (H2) that astrophysical collapse produces genuine event horizons rather than ultra-compact horizonless mimickers; and (H3) that the observed black-hole population is produced predominantly by stellar core collapse followed by binary and hierarchical mergers. A web-based literature survey across primary journals, collaboration releases, and living reviews identifies four research gaps that motivate these hypotheses. The verification plan combines black-hole spectroscopy of gravitational-wave ringdowns, event-horizon-scale imaging consistency tests, stacked searches for post-merger echoes and tidal deformability, and hierarchical Bayesian population inference on the LIGO–Virgo–KAGRA catalogs, with explicit decision rules, a 36-month timeline, and a risk register. Expected results range from percent-level tests of the Kerr hypothesis to quantitative formation-channel evidence, with a predefined protocol for falsification.

Index Terms—black holes, singularity theorems, gravitational collapse, gravitational waves, black-hole spectroscopy, Event Horizon Telescope, research plan.

I. INTRODUCTION

Why do black holes exist? The short answer taught in modern astrophysics is that black holes form when sufficiently massive stars burn out and collapse in on themselves: once nuclear fusion has assembled an iron core, no further exothermic burning is possible, and if the remnant exceeds the maximum mass supportable by neutron-degeneracy pressure, nothing halts the contraction. What elevates this picture from a heuristic argument to a theorem is the work of Roger Penrose, who in January 1965 proved that, within general relativity (GR), the formation of a singularity is *generic* once a trapped surface forms during collapse [1]. For “the discovery that black hole formation is a robust prediction of the general theory of relativity,” Penrose was awarded one half of the 2020 Nobel Prize in Physics [2].

Half a century after the theorem, the observational situation has been transformed. The Laser Interferometer Gravitational-Wave Observatory (LIGO) and Virgo detected GW150914, the merger of two stellar-mass black holes [11]; the Event Horizon Telescope (EHT) imaged the shadows of the supermassive black holes M87* and Sagittarius A* [18], [19]; and decades of stellar-orbit monitoring established a $\sim 4 \times 10^6 M_\odot$ dark mass at the Milky Way’s center [20], [21]. Two clarifications of terminology are worth making. First, the progenitors of stellar-mass black holes are *massive* stars (zero-age main-sequence masses $\gtrsim 20\text{--}25 M_\odot$, depending on metallicity and rotation), not “supermassive” ones; the latter term is reserved in the literature for the $10^5\text{--}10^{10} M_\odot$ black holes in galactic nuclei, whose origin is a distinct and still open problem. Second, Penrose’s theorem proves geodesic incompleteness under stated assumptions; it does not by itself establish that the resulting objects are the Kerr black holes of astrophysics, nor that their horizons are observationally distinguishable from exotic alternatives.

This document therefore treats the continued question “why do black holes exist?” as an *empirical research program*: to what precision do astrophysical compact objects behave as the black holes predicted by GR? We formulate three research questions:

- **RQ1.** Are the compact remnants of gravitational collapse consistent with the Kerr metric and its no-hair property, and to what precision?
- **RQ2.** Do astrophysical compact objects possess event horizons (as expected from weak cosmic censorship), or can horizonless ultra-compact mimickers be detected or excluded?
- **RQ3.** Do stellar core collapse and subsequent mergers account quantitatively for the observed black-hole population, or must additional formation channels be invoked?

The contributions of this plan are: (i) a structured web-based literature survey with an explicit protocol (Section II); (ii) a gap analysis distilled into four research gaps (Section III); (iii) three falsifiable candidate hypotheses mapped to observables and datasets (Section IV); (iv) a verification design with

decision rules, timeline, and risk register (Section V); and (v) expected outcomes under confirmation and falsification branches (Section VI).

II. LITERATURE REVIEW

A. Survey Protocol

A structured web survey was performed in August 2026. Sources searched included journal platforms (APS Physical Review Letters/X/D, *The Astrophysical Journal Letters*, *Science*, Springer’s *Living Reviews in Relativity* and *The Astronomy and Astrophysics Review*), collaboration portals (LIGO/Virgo/KAGRA GWOSC, EHT), the Nobel Foundation archive, and bibliographic indexes (OpenAlex, Semantic Scholar) for metadata verification. Queries combined historical terms (“Penrose singularity theorem,” “Oppenheimer–Snyder collapse”), observational terms (“GW150914,” “GWTC catalog,” “EHT shadow,” “ringdown overtones”), and frontier terms (“cosmic censorship status,” “exotic compact objects,” “black hole seed formation”). Inclusion criteria were: peer-reviewed primary articles or collaboration papers, authoritative reviews, or official releases; English; no date limit for foundational works, with a preference for 2019–2026 literature on observational tests. Approximately 45 records were screened and 33 retained; they are cited throughout and listed in the References.

B. Theoretical Foundations: From Exact Solutions to Collapse

Einstein’s field equations admitted black-hole-type solutions almost immediately: Schwarzschild’s spherically symmetric vacuum solution (1916) contains the gravitational radius [4]

$$r_s = \frac{2GM}{c^2}, \quad (1)$$

and Kerr’s rotating solution (1963) supplied the metric of realistic, astrophysical black holes [5]. Whether these solutions described real objects was debated for decades; notably, Einstein himself argued in 1939 that a cluster of orbiting particles could not be compressed below the Schwarzschild radius, concluding that “Schwarzschild singularities do not exist in physical reality” [8].

The case for stellar collapse was assembled in three steps. Chandrasekhar showed that relativistic electron degeneracy cannot support white dwarfs above $\approx 1.4 M_\odot$ [6]. Oppenheimer and Volkoff then derived the general-relativistic maximum mass of neutron cores [7], and Oppenheimer and Snyder modeled the complete collapse of a pressureless dust sphere: to a distant observer the surface asymptotically freezes near $r = 2GM/c^2$, while the comoving proper time to the singularity is finite [3]. The modern value of the maximum neutron-star mass is $\approx 2\text{--}3 M_\odot$ depending on the dense matter equation of state; compact remnants above this limit have no known equilibrium and are black-hole candidates.

C. Penrose’s Theorem and the 2020 Nobel Prize

The decisive step replaced exact integration with global, causal methods. Penrose’s 1965 theorem states that if (i)

the spacetime possesses a non-compact Cauchy surface, (ii) the null energy condition holds, and (iii) a closed trapped surface forms—a two-surface on which both future-directed null expansions are negative—then the spacetime is future null geodesically incomplete [1]:

$$\theta_+ < 0, \theta_- < 0 \implies \text{incomplete future null geodesics.} \quad (2)$$

Two features make the result profound. First, no symmetry is assumed, so the conclusion applies to generic, non-spherical collapse—unlike the idealized Oppenheimer–Snyder model. Second, the theorem proves *incompleteness*—the rigorous content of “singularity”—rather than a curvature divergence per se. Hawking and Penrose generalized the result in 1970 to a family of conditions covering both collapse and cosmology [9]. Penrose also formulated the cosmic censorship conjecture in 1969, proposing that singularities formed in generic collapse are hidden behind event horizons [10]. The 2020 Nobel Prize in Physics credited precisely this body of work: one half to Penrose, one half jointly to Genzel and Ghez for the discovery of the supermassive compact object at the Galactic center [2].

D. Observational Pillars

1) *Gravitational waves*: GW150914, detected on 14 September 2015 with a false-alarm probability corresponding to $> 5\sigma$, was the merger of two black holes of ≈ 36 and $\approx 29 M_\odot$, leaving a $\approx 62 M_\odot$ remnant and radiating $\approx 3 M_\odot c^2$ in gravitational waves [11]. The GWTC-3 catalog reports 90 confident compact-binary mergers through the third observing run, of which roughly 88 are binary-black-hole (BBH) systems, with source-frame component masses from ~ 5 to $\sim 105 M_\odot$ and generally modest effective spins [12]; subsequent observing runs have extended the confident event list to well over one hundred [13]. Black-hole spectroscopy—fitting the ringdown as a sum of quasi-normal modes (QNMs),

$$h(t) = \sum_{\ell mn} A_{\ell mn} e^{-t/\tau_{\ell mn}} \cos(\omega_{\ell mn} t + \phi_{\ell mn}), \quad (3)$$

whose frequencies $\omega_{\ell mn}$ and damping times $\tau_{\ell mn}$ depend only on the remnant’s mass and spin if it is a Kerr black hole—produced a claimed 3.6σ detection of the $\ell = m = 2$ fundamental mode plus one overtone in GW150914 [14], [15]. This claim was subsequently challenged by an independent analysis attributing it to ringdown start-time systematics [16], with a reply defending the original methodology [17]; the latest living review treats the overtone status as unresolved [25].

2) *Horizon-scale imaging*: The EHT resolved the lensed photon ring and shadow of M87*: a ring of diameter $42 \pm 3 \mu\text{as}$ around a $(6.2\text{--}6.5) \times 10^9 M_\odot$ black hole, consistent with Kerr predictions [18]. The Sgr A* image shows a ring of diameter $51.8 \pm 2.3 \mu\text{as}$, against a GR prediction of $52 \pm 0.4 \mu\text{as}$ anchored by stellar-orbit mass measurements [19], [23]. However, the same review stresses that present images do not by themselves constitute direct horizon tests: the arguments depend on emission-model assumptions and on $\sim 20 \mu\text{as}$ resolution [23].

TABLE I
RESEARCH GAPS IDENTIFIED BY THE LITERATURE SURVEY.

ID	Gap	Key evidence
G1	Kerr/no-hair property tested only at the $\sim 10\%$ level; ringdown overtone claims disputed due to start-time systematics.	[14], [16], [17], [25]
G2	No decisive test of horizons; exotic compact objects (ECOs) excluded only partially; weak cosmic censorship unproven, strong censorship false in its C^0 form, naked singularities constructible.	[24], [29]–[32]
G3	Formation channels: GW190521’s components lie in the pair-instability mass gap; seed mechanisms for high-redshift supermassive black holes cannot yet explain observations without fine-tuned abundances.	[26], [27]
G4	Singularities and information loss remain theoretically unresolved (context; out of direct empirical scope).	[28], [33]

3) *Dynamical mass measurements*: Stellar orbits establish $\approx 4 \times 10^6 M_\odot$ within a Solar-System-sized volume at Sgr A* [20], [21]. For stellar-mass objects, black-hole candidates are identified in X-ray binaries when dynamical modeling yields a compact-object mass above the maximum neutron-star mass; the archetype Cygnus X-1 contains a $21.2 \pm 2.2 M_\odot$ black hole [22].

III. CONCLUSIONS OF THE SURVEY AND RESEARCH GAPS

Four conclusions emerge. First, the *existence* of black holes, in the operational sense of compact objects with masses above the neutron-star limit and no detectable surface, is established beyond reasonable doubt by mutually independent channels. Second, all current data are consistent with the Kerr paradigm—but at modest precision. Third, on the theory side the two deepest questions raised by Penrose’s theorem, cosmic censorship and the nature of the singularity, remain open. Fourth, formation-channel questions are empirically unresolved precisely where stellar-evolution theory predicts structure (the pair-instability mass gap) and where cosmology demands early growth (high-redshift quasars).

These conclusions crystallize into four gaps, summarized in Table I.

G1 (precision of the Kerr hypothesis). Black-hole spectroscopy of GW150914 probed the no-hair property at only the $\sim 10\%$ level, and the status of the claimed first overtone remains contested [16], [17], [25]. EHT shadow sizes agree with Kerr predictions but are limited by resolution and emission-model dependence [23]. A systematic, start-time-robust spectroscopy program on the growing catalog is missing.

G2 (do horizons exist?). In GR, stationary black holes have vanishing tidal Love numbers and no reflective surface,

whereas ECOs such as gravastars or boson stars predict echoes after the merger and nonzero multipole structure [24]. Searches so far have returned null results that “strengthen and quantify the Kerr paradigm” without excluding all mimickers [24]. Theoretically, weak cosmic censorship has been proven only in restricted matter models; strong cosmic censorship fails in its original C^0 formulation for Kerr interiors [29], [30]; naked singularities have been constructed in scalar-field collapse and, more recently, for the vacuum equations [31], [32]. Whether nature censors singularities is thus both mathematically and observationally open.

G3 (formation channels). GW190521 merged black holes of ≈ 85 and $\approx 66 M_\odot$ —inside or adjacent to the pulsational pair-instability mass gap predicted by stellar evolution—challenging isolated single-star channels [26]. At the other extreme, high-redshift quasars and JWST-era black holes strain seed models: light (Pop III remnant) seeds must outnumber heavy (direct-collapse) seeds by 10^3 – 10^5 , yet the rare heavy channels most easily explain the earliest giants; the seed problem “remains an unsolved problem” [27].

G4 (interiors and quantum theory; context). Semiclassical analysis yields Hawking radiation at

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}, \quad (4)$$

and an apparent information paradox [28]; recent quantum information formulations sharpen the problem but leave the microscopic mechanism open [33]. Because interiors are causally sealed, G4 is documented for completeness but excluded from the empirical scope of this plan.

IV. CANDIDATE HYPOTHESES

Each hypothesis is stated in falsifiable form, mapped to observables, data, and a decision criterion in Table II and Fig. 1.

A. H1: The Kerr Remnant Hypothesis

The compact object produced by stellar collapse is described externally by the Kerr metric. Equivalently, its multipole moments obey

$$M_\ell + iS_\ell = M (i\chi)^\ell, \quad (5)$$

with mass M and dimensionless spin χ , and its QNM spectrum (3) is fixed entirely by (M, χ) . H1 is falsified by a persistent, instrumentally unexplained inconsistency among independently measured QNM frequencies, or by a shadow size incompatible with the mass-and-distance-anchored Kerr prediction.

B. H2: The Horizon Hypothesis

Generic collapse produces an event horizon: observed compact objects are not horizonless ultra-compact objects. We parameterize departures by the ECO compactness $\varepsilon = 1 - r_s/R$ [24]. H2 is falsified by statistically significant post-merger echoes consistent across detectors, or by a measured nonzero tidal Love number.

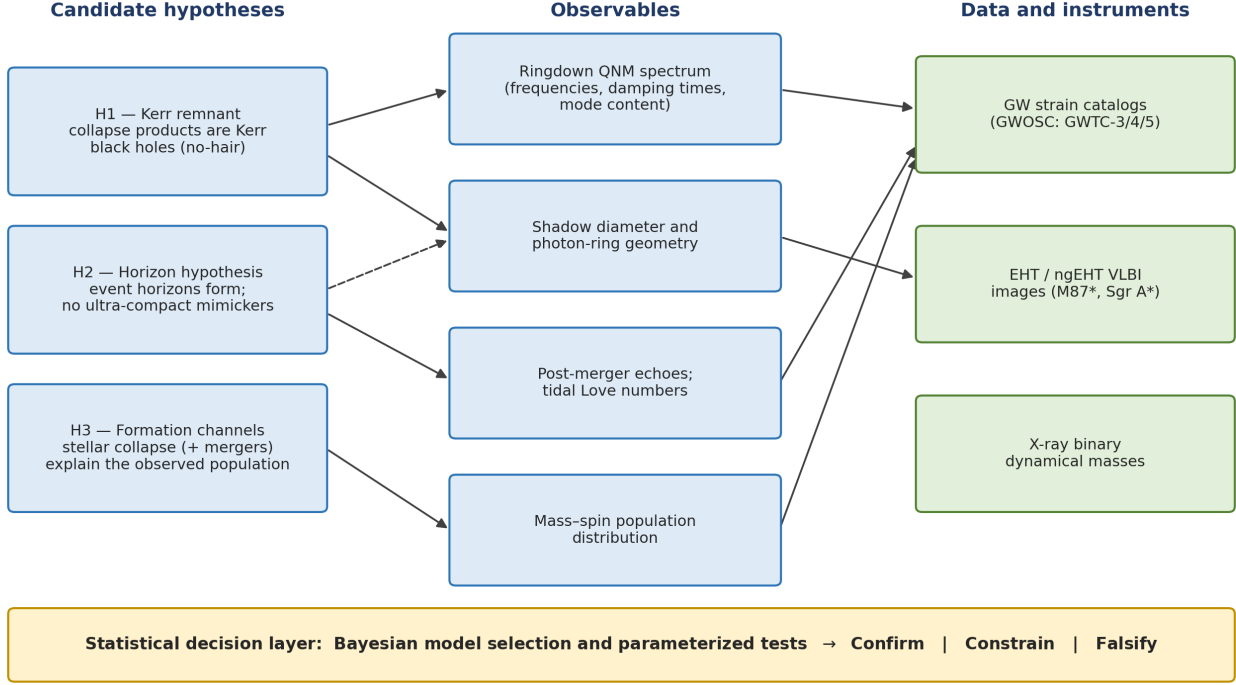


Fig. 1. Research framework: candidate hypotheses are mapped to observables and thence to existing public data; a statistical decision layer returns confirmation, quantitative constraints, or falsification.

TABLE II
MAPPING OF HYPOTHESES TO OBSERVABLES, DATASETS, AND DECISION CRITERIA.

Hypothesis	Observables	Data	Decision criterion
H1: Kerr remnant (no-hair)	QNM frequencies/damping times of two or more modes; shadow diameter; Fe $K\alpha$ profiles	GWTC-3/4/5 ringdown events with $\text{SNR}_{\text{ring}} \geq 20$; EHT M87*, Sgr A*	Kerr rejected only if $\ln B_{\text{nonKerr}} > 8$ in ≥ 2 independent events, or $ \delta\hat{\omega} , \delta\hat{\tau} > 5\%$ persistently
H2: Horizons (no mimickers)	Post-merger echo trains; tidal Love numbers; inspiral tidal deformability	Stacked GWTC events; BNS/NSBH inspirals	Horizonless models with compactness $\varepsilon < 10^{-4}$ excluded at 90% credibility, or echo energy fraction η upper limit below 10^{-3}
H3: Formation channels	Component mass-spin population; mass-gap subpopulation fraction; merger-rate evolution	GWTC catalogs; COMPAS-style population synthesis	Channel model rejected if hierarchical Bayes factor $\ln B < -8$; primordial fraction constrained to $f_{\text{PBH}} < f_{\text{max}}$

C. H3: The Formation-Channel Hypothesis

The observed BBH population arises predominantly from core collapse of massive stars in binaries, processed through isolated evolution and dynamical or hierarchical mergers, with no primordial component required above a small fraction f_{max} . H3 is falsified if no stellar-evolution-consistent population model fits the mass spectrum—in particular the pair-instability gap region populated by GW190521-like events [26]—without a substantial non-stellar component.

V. VERIFICATION PLAN

A. Data, Software, and Infrastructure

All primary data are public: strain data and parameter estimates from the GWOSC event portal [13]; EHT published

data products [18], [19]; and published X-ray binary dynamical masses [22]. Analysis will use established open-source infrastructure for Bayesian parameter estimation and population inference, frequency- and time-domain ringdown likelihoods, numerical-relativity surrogate waveforms for injection studies, and containerized pipelines to guarantee reproducibility. Compute requirements (Markov-chain and nested sampling campaigns) are modest by current standards and schedulable on institutional clusters.

B. Testing H1: Robust Black-Hole Spectroscopy

Event selection targets BBH mergers with ringdown signal-to-noise ratio $\text{SNR}_{\text{ring}} \geq 20$, expected several times per year at O4/O5 sensitivity. For each event we fit multitone models ($\ell = m = 2$, $n = 0, 1, 2$; and $\ell = m = 3$ where

supported) and explicitly address the start-time controversy [16], [17] by (i) scanning the ringdown start time over a physically motivated window and reporting the stability of the Bayes factor, (ii) repeating the analysis in both time and frequency domains, and (iii) validating the pipeline on numerical-relativity injections where the true mode content is known. Parameterized deviations $\delta\hat{\omega} = (\omega - \omega_{\text{Kerr}})/\omega_{\text{Kerr}}$ and $\delta\hat{\tau}$ are estimated jointly for all modes; consistency of two modes at the same (M, χ) constitutes the no-hair test. Results are combined across events hierarchically. Decision rule: H1 stands unless the non-Kerr model is strongly favored ($\ln B > 8$) in at least two independent events, or deviations exceed 5% persistently—in which case systematic checks (calibration, noise transients) precede any physical interpretation. EHT shadow measurements of Sgr A* and M87*, anchored by independent mass and distance estimates [23], and Fe K α reflection fits in bright X-ray binaries, provide cross-checks on complementary systematics.

C. Testing H2: Echoes and Tidal Response

A stacked, template-agnostic search for post-merger echoes is performed over GWTC events using time-delay combs set by the light-travel time of a near horizon cavity, with echo waveform templates spanning the main ECO families [24]. Controls include time-reversed data, off-source trials, and per-event noise consistency tests to guard against the false positives that have affected this field. In parallel, inspiral tidal deformability measurements from BNS and NSBH events constrain the Love numbers of the lighter companions, while future high-SNR BBH inspirals bound the deformability of putative massive ECOs. Decision rule: absence of echoes at stacked SNR > 8 translates into upper limits on the echo energy fraction η and excludes horizonless models with $\varepsilon < 10^{-4}$ at 90% credibility; any positive detection triggers the systematics protocol before interpretation.

D. Testing H3: Population Inference

A hierarchical Bayesian analysis infers the hyperparameters Λ of the BBH population model from all catalog events,

$$p(d | \Lambda) = \prod_i \frac{\int p(d_i | \theta_i) p(\theta_i | \Lambda) d\theta_i}{\pi(\theta_i)}, \quad (6)$$

with selection effects computed by Monte Carlo injection campaigns. Competing formation scenarios—isolated binary evolution, dynamical assembly in clusters, hierarchical mergers in active galactic nuclei, and a primordial component—are forward-modeled with population-synthesis codes and compared via Bayes factors. Special attention is given to the pair-instability mass gap: the fraction of events in the gap region and their spin distribution discriminate between modified stellar physics, hierarchical growth, and non-stellar origins [26]. For supermassive black holes, the plan limits itself to synthesizing seed-model predictions against quasar luminosity functions, given the open state of that problem [27].

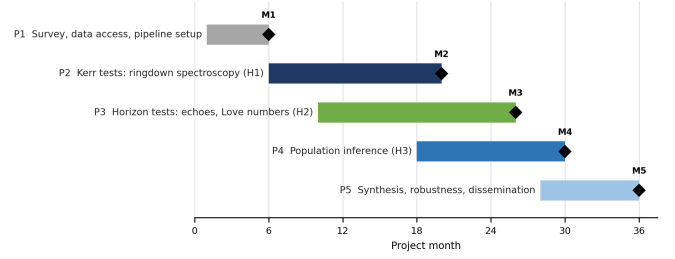


Fig. 2. Project schedule: five phases with milestones M1–M5 (diamonds).

TABLE III
MILESTONES AND DELIVERABLES.

Month	Milestone	Deliverable
6	M1	Analysis pipelines validated on public data and injections
20	M2	H1 spectroscopy results (catalog-level) as preprint
26	M3	H2 echo/Love-number constraints released
30	M4	H3 population-model comparison with Bayes factors
36	M5	Integrated results, open-source release, final report

E. Validation and Reproducibility

Three safeguards are built in. First, blind injection studies: simulated Kerr-violating signals and synthetic echoes are inserted into real noise to measure recovery fidelity and false-alarm rates before unblinding. Second, internal red-team review of any claimed deviation prior to publication. Third, all pipelines, configuration files, and analysis products are released publicly with the manuscripts.

F. Timeline and Milestones

The plan spans 36 months in five overlapping phases (Fig. 2, Table III).

G. Risks and Mitigations

Table IV lists principal risks. The recurring theme is that claimed deviations in black-hole tests have historically traced to systematics (start-time choices, calibration, emission modeling), so every falsification path includes a mandatory systematics protocol.

VI. EXPECTED RESULTS

Under the confirmation branch, the plan expects to sharpen the Kerr hypothesis tests from the current $\sim 10\%$ level [25] toward the few-percent level for $\delta\hat{\omega}$ and $\delta\hat{\tau}$ by stacking several high-SNR events, to push ECO compactness exclusions to $\varepsilon \lesssim 10^{-4}$ for the main mimicker families, and to deliver the first catalog-scale Bayes-factor comparison of stellar formation channels including an explicit mass-gap subpopulation fraction. These results would convert “black holes exist” from a qualitative statement into a quantified one, with error bars.

TABLE IV
RISK REGISTER.

Risk	Mitigation
Ringdown start-time and model systematics [16]	Start-time scans; dual time/frequency analyses; NR injection validation
EHT emission-model dependence [23]	Use shadows only as cross-check; require mass/distance anchoring
Catalog revisions and selection effects	Re-run population inference on each catalog release; explicit selection modeling (6)
Echo false positives	Time-reversed and off-source controls; blind analysis
Compute budget over-runs	Surrogate waveforms; event subselection by SNR; cluster scheduling

Under the falsification branch—persistent QNM inconsistency, a robust echo detection, or an unmodelable mass-gap population—the plan yields parameterized departures from Kerr, an ECO compactness measurement, or a required non-stellar fraction f_{PBH} , each with a predefined systematics protocol to establish credibility before interpretation. Either branch produces publishable, reproducible results; the decision rules of Section V prevent post-hoc narrative fitting.

Beyond the hypotheses, expected deliverables include an open ringdown spectroscopy pipeline with start-time-robustness reporting, a stacked echo search framework with calibrated false-alarm rates, three to four peer-reviewed articles, and training of graduate students in gravitational-wave data analysis.

VII. CONCLUSION

Penrose answered “why do black holes exist?” theoretically: once a trapped surface forms in collapsing matter, singularities—and with them black holes—are generic predictions of general relativity [1], a result honored with the 2020 Nobel Prize [2]. This research plan carries the question forward empirically. Three falsifiable hypotheses distilled from a structured literature survey target the precision of the Kerr description, the reality of horizons, and the stellar origin of the observed population. With public data, established statistical machinery, explicit decision rules, and a 36-month schedule, the plan is designed to produce quantitative answers—or, with equal success, the first credible evidence that nature deviates from the black holes Einstein’s equations predicted and Penrose proved must form.

APPENDIX A SURVEY SEARCH LOG

Representative queries executed during the August 2026 web survey, with the databases used:

- “Penrose singularity theorem 1965 trapped surface” — APS, Nobel Foundation archive, OpenAlex.
- “Oppenheimer–Snyder gravitational collapse 1939,” “Einstein black holes prediction doubt” — APS, Annals of Mathematics records.

- “GW150914 first gravitational wave detection,” “GWTC-3 catalog binary black hole population” — GWOSC event portal, Phys. Rev. X.
- “Event Horizon Telescope M87 shadow 2019,” “Sgr A* EHT image 2022” — EHT/ESO announcements, ApJ Letters.
- “ringdown overtone GW150914,” “black hole spectroscopy tests of general relativity” — OpenAlex meta-data verification (Cotesta et al. PRL 129, 111102, 2022; Isi and Farr PRL 131, 169001, 2023).
- “cosmic censorship conjecture status counterexample,” “exotic compact objects gravastar constraints” — Living Reviews in Relativity, Comptes Rendus Mécanique.
- “supermassive black hole seed formation direct collapse problem,” “pair-instability mass gap LIGO” — Open Journal of Astrophysics, ApJ Letters.

Access date for all online sources: August 30, 2026.

APPENDIX B KEY RELATIONS

For reference, the principal relations used in the plan: the gravitational radius (1); the Chandrasekhar mass scale $M_{\text{Ch}} \propto (\hbar c/G)^{3/2} m_p^{-2} \approx 1.4 M_{\odot}$; the trapped-surface criterion (2); the ringdown mode expansion (3); the no-hair multipole relation (5); the ECO compactness $\varepsilon = 1 - r_s/R$; the Hawking temperature (4); and the hierarchical population likelihood (6).

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